Program at a glance (IEEE ai4i and IEEE AIKE 2018)

September 26 (Wednesday)

Place / Time	Crystal Ballroom	Platinum Room	Crystal 3	Slate Room	Veranda	Mezzanine
07:30-17:00	Registration					
07:30-08:20					Breakfast	
08:40-09:00	**Opening Ceremony**					
09:00-09:50	**Keynote I – AIKE** Nik Bourbakis					
09:50-10:10	Coffee Break					
10:10-11:00	**Keynote II – ai4i** David Newman					
11:10-12:00	**Keynote III – AIKE** Bhavani Thuraisingham					
12:00-13:00	Lunch (on your own)					
13:00-14:50	AIKE(1) – Machine Learning (1+3 papers)	**ai4i(1)** (4 papers - **best**)				
14:50-15:10	Coffee Break					
15:10-16:50	AIKE(2) – Deep Learning (2+2 papers)	**ai4i(2)** (5 papers)	**AIKE Workshops** **(AI3D - 4 papers)**			
16:50-18:10	AIKE(3) – Best AIKE18 Papers (3 papers)	**ai4i(3)** (4 papers)				
18:00-19:00	**ai4i planning session**					
19:30-21:30	Reception					

September 27 (Thursday)

Place / Time	Crystal Ballroom	Platinum Room	Crystal 3	Slate Room	Veranda	Mezzanine
07:30-17:00	Registration					
07:30-08:30					Breakfast	
08:30-09:20	**Keynote IV – AIKE** Carlo Zaniolo					
09:30 -10:20	**Keynote V – ai4i** Nestor Rychtyckyj					
10:20-10:40	Coffee Break					
10:40 - 11:30	**Keynote VI – AIKE** Sang-goo Lee					
11:40 – 12:30	**Keynote VII – AIKE** Ophir Frieder					
12:30-13:30	Lunch (on your own)					

13:30-15:20	AIKE(4) – Knowledge Graph Representations and Search (2+4 papers)	**Tutorials – ai4i**	**AIKE Workshops (ABC - 6 papers)**			
15:20-15:30	Coffee Break					
15:30-16:30	AIKE(5) – Ensemble Learning (3 papers)	**Tutorials – ai4i**	**AIKE Workshops (ABC - 4 papers)**			
16:30-16:40	Intermission					
16:40-18:00	AIKE(6) – Poster Session (9 poster papers)	**Tutorials – ai4i**	**AIKE Workshops (IDP - 6 papers)**			
18:00-19:00	**AIKE planning session**					
19:30-21:30	**Banquet** Keynote: Raymond Yeh					

September 28 (Friday)

Place / Time	Crystal Ballroom	Platinum Room	Crystal 3	Slate Room	Veranda	Mezzanine
07:30-17:00	Registration					
07:30-09:00					Breakfast	
09:00-09:50	**Keynote VIII - ai4i** Ravi Iyer					
09:50-10:10	Coffee Break					
10:10-11:00	**Keynote IX – AIKE** Paul Conoval					
11:10 – 12:00	**AIKE distinguished tutorial** Oscar Garcia					
12:00-13:00	Lunch (on your own)					
13:00-15:00	AIKE(7) – AI methods for Sentiment and social Analysis (4+2 papers)	**ai4i(4)** (4 papers)	**ai4i(5)** (4 papers)			
15:00-15:20	Coffee Break					
15:20-17:20	AIKE(8) – Innovative Approaches in AI and KE (3 short papers)	**ai4i(6)** (4 papers)	**ai4i(7)** (4 papers)			

Detailed Sessions (IEEE ai4i 2018)

Session	Title
ai4i(1) – Best Paper Session Session Chair:	Nikhil Nigam, Sanjay Lall, Pedram Hovareshti, Kristopher Ezra, Linas Mockus, Devendra Tolani and Shawn Sloan. *Sufficient Statistics for Optimal Decentralized Control in System of Systems*
	Guangzhi Qu. *Genetic Algorithm based Parallelization Planning for Legacy Real-Time Embedded Programs*
	Jonas Stricker, Benno Koeppl, Andi Buzo, Jerome Kirscher, Linus Maurer and Georg Pelz. *Efficient simulative Pass/Fail Characterization applied to automotive Power Steering*
	Athanasios I. Kyritsis, Geoffrey Willems, Michel Deriaz and Dimitri Konstantas. *Gait Recognition with Smart Devices Assisting Postoperative Rehabilitation in a Clinical Setting*
ai4i(2) – Data Fusion, Data Compression and Algorithms Session Chair:	Jun Jo, Yong Oh Lee and Jongwoon Hwang. *Multi-layer Nested Scatter Plot: A data wrangling method for correlated multi-channel time series signals*
	Brian Beaton. *The Moat Effects of Data Swamps*
	Marcio Moreno, Rodrigo C. M. Santos, Reinaldo Mozart, Wallas H. S. Dos Santos and Renato Cerqueira. *Assisting Seismic Image Interpretations through Hyperknowledge*
	Yu-Wei Wen and Chuan-Kang Ting. *Designing Facility Layouts with Hard and Soft Constraints by Evolutionary Algorithm*
	Jake Fitzsimmons and Pablo Moscato. *Symbolic Regression Modeling of Drug Responses*
ai4i(3) – Machine Learning Session Chair:	Zhu Zhou and Tong Zhang. *Applying Machine Learning to Service Assurance in Network Function Virtualization Environment*
	James Obert and Tom Mannos. *Semi-Supervised Learning and ASIC Path Verification*
	Xinyuan Huang, Amit Saha, Debojyoti Dutta and Ce Gao. *Kubebench: A Benchmarking Platform for ML Workloads*
	Debojyoti Dutta, Amit Saha, Johnu George, Xinyuan Huang, Ramdoot Pydipaty, Purushotham Kamath and Lew Tucker. *Towards #consistentAI*
ai4i(4) – Frameworks and Predictions Session Chair:	Ayhan Demiriz. *Used Car Pricing and Beyond: A Survival Analysis Framework*
	Rua Alsuroji, Nizar Bouguila and Nuha Zamzami. *Predicting defect-prone software modules using shifted-scaled Dirichlet distribution*
	Nuha Zamzami and Nizar Bouguila. *Consumption Behavior Prediction using Hierarchical Bayesian Frameworks*
	Daniela D'Auria and Fabio Persia. *Design of a Framework allowing Researchers to Optimize their Academic Evaluation*
ai4i(5) – Cyber Physical Systems Session Chair:	Manuel Roveri, Dario Cogliati, Mirko Falchetto, Danilo Pau and Gabriele Viscardi. *Intelligent Cyber-Physical Systems for Industry 4.0*
	Jérôme Mendes, Ricardo Maia, Rui Araújo and Gonçalo Gouveia. *Intelligent Controller for Industrial Processes Applied to a Distributed Two-Tank System*
	Liu Chia-Ruei, Chen Po-Wei, Yang Chao-Chun and Duan Li-Hua. *Monitoring Machine Tool Based On External Physical Characteristics Of The Machine Tool Using Machine Learning Algorithm*
	Florian Jaensch, Akos Csiszar, Annika Kienzlen and Alexander Verl. *Reinforcement Learning of Material Flow Control Logic using Hardware-in-the-Loop simulation*
ai4i(6) – Artificial Intelligence Session Chair:	Louis Rosenberg, David Askay, Lynn Metcalf, Erick Harris and Gregg Willcox. *Amplifying the Social Intelligence of Teams Through Human Swarming*
	Vinod Muthusamy, Aleksander Slominski and Vatche Ishakian. *Towards enterprise-ready artificial intelligence deployments*

	Stephan Matzka. *Using Process Quality Prediction to increase Resource Efficiency in Manufacturing Processes*
	Gian Antonio Susto, Matteo Terzi, Chiara Masiero, Simone Pampuri and Andrea Schirru. *A Fraud Detection Decision Support System via Human On-line Behavior Characterization and Machine Learning*
ai4i(7) – Neural Network Session Chair:	Marcelo Nery, Marcio Moreno, Rodrigo C. M. Santos and Wallas H. S. Dos Santos. *Facing Digital Agriculture Challenges with Knowledge Engineering*
	Ryota Shimizu, Kosuke Asako, Hiroki Ojima, Shohei Morinaga, Mototsugu Hamada and Tadahiro Kuroda. *Balanced Mini-batch Training for Imbalanced Image Data Classification with Neural Network*
	Guangzhi Qu. *Detection Sound Source Direction in 3D Space Using Convolutional Neural Networks*
	Javad Rouzafzoon and Petri Helo. *Developing Logistics and Supply Chain Management by Using Agent-Based Simulation*

Detailed Sessions (IEEE AIKE 2018)

Session	Title
AIKE(1) – Machine Learning Session Chair:	38 Ganapathy Mani, Bharat Bhargava, Pelin Angin, Miguel Villarreal-Vasquez, Denis Ulybyshev and Jason Kobes. Machine Learning Models to Enhance the Science of Cognitive Autonomy
	17 Chieh-Kang Liao, Alan Liu and Yu-Sheng Chao. A Machine Learning Approach to Case Adaptation
	48 Nikki Lijing Kuang and Clement H. C. Leung. Performance Dynamics and Termination Errors in Reinforcement Learning
AIKE(2) – Deep Learning Session Chair:	14 Shin Kamada, Takumi Ichimura and Toshihide Harada. Adaptive Structural Learning of Deep Belief Network for Medical Examination Data and Its Knowledge Extraction by using C4.5
	26 Pengwei Hu, Zhaomeng Niu, Tiantian He and Keith C.C. Chan. Learning Deep Representations in Integrated Large Network for Graph Clustering
	39 Ganapathy Mani, Bharat Bhargava and Jason Kobes. Scalable Deep Learning Through Fuzzy-based Clustering in Autonomous Systems
AIKE(3) – Best AIKE18 Papers Session Chair:	21 Somang Nam and Deborah Fels. Assessing closed captioning quality using a multilayer perceptron
	23 Shah Muhammad Hamdi, Berkay Aydin, Soukaina Filali Boubrahimi, Rafal Angryk, Lisa Crystal Krishnamurthy and Robin Morris. Biomarker Detection from fMRI-based Complete Functional Connectivity Networks
	45 Mattia Atzeni and Maurizio Atzori. Translating Natural Language to Code: an Unsupervised Ontology-based Approach
AIKE(4) – Knowledge Graph Representations and Search Session Chair:	25 Ying Fung Yiu, Jing Du and Rabi Mahapatra. Evolutionary Heuristic A* search: Heuristic Function Optimization via Genetic Algorithm
	20 Graph-based Ontology Summarization: A Survey
	36 Gustavo De Assis Costa and José Maria Parente Oliveira. Towards Exploring Literals to enrich Data Linking in Knowledge Graphs
	46 Ajay Kshemkalyani and Faizan Ali. Fast Graph Exploration by a Mobile Robot
	33 Marcio Moreno, Rodrigo C. M. Santos, Wallas H. S. Dos Santos, Reinaldo Silva and Renato Cerqueira. Knowledge Bases Enrichment with Temporal Reasoning using Hyperknowledge
	22 Angelina A. Tzacheva, Arunkumar Bagavathi and Sharath C.B. Suryanarayanaprasad. In Search of Actionable Patterns of Lowest Cost - a Scalable Action Graph Method
AIKE(5) – Ensemble Learning Session Chair:	54 Justin Kehl and Lubomir Stanchev. N-SLOPE: A One-Class Classification Ensemble For Nuclear Forensics
	19 Soukaina Filali Boubrahimi, Ruizhe Ma, Berkay Aydin and Rafal Angryk. Neuro-Ensemble
	6 Nouna Khandan and Ali Ghobadi. A Comprehensive Approach to Integrate Multiple Fuzzy Expert Systems
AIKE(6) – Poster Session Session Chair:	24 Soukaina Filali Boubrahimi and Rafal Angryk. Multivariate Time Series Nearest Neighbor Search: A Case Study on Solar Flare Prediction
	53 Emmanuel Oyekanlu and Kevin Scoles. Towards Low-Cost, Real-Time, Distributed Signal and Data Processing for Artificial Intelligence Applications at Edges of Large Industrial and Internet Networks

	35 Bogdan Walek and Petra Spackova. Content-based recommender system for online stores using expert system
	11 Wedad Alawad, Mohamed Zohdy and Debatosh Debnath. Tuning Hyperparameters of Decision Tree Classifiers using Computationally Efficient Schemes
	12 Pablo Andrés Millan Arias and Julián Armando Quiroga Sepúlveda. Deep Learned vs. Hand-Crafted Features for Action Classification
AIKE(7) – AI methods for Sentiment and social Analysis Session Chair:	15 Dongyang Liang, Shasha Li, Bing Ji, Zibo Yi, Jie Yu, Jianfeng Zhang, Yusong Tan and Qingbo Wu. Research on Matrix Multiplication Based WeChat Group Tagging Technology
	37 Romita Banerjee, Karima Elgarroussi, Sujing Wang, Yongli Zhang and Christoph F. Eick. Tweet Emotion Mapping: Understanding US Emotions in Time and Space
	13 Yongli Zhang and Christoph F. Eick. A Novel Two-stage System for Detecting and Tracking Events in Twitter
	4 Jaishree Ranganathan, Nikhil Hedge, Allen Irudayaraj and Angelina Tzacheva. Automatic Detection of Emotions in Twitter Data - A Scalable Decision Tree Classification Method
	30 Minh-Son Dao, Koji Zettsu and Asem Kasem. A Raster-Image-Based Approach for Understanding Associations of Urban Sensing Data
AIKE(8) – Innovative Approaches in AI and KE Session Chair:	*43 Yu-Heng Hsieh and Ming-Syan Chen. Toward green computing: striking the trade-off between efficiency and energy consumption of parallel sequential pattern mining on GPU*
	61 Markus Lemmer, Florian Köpf, Stefan Schwab, Michael Flad and Sören Hohmann. Modeling of Human-Centered Cooperative Control by Means of Tracking in Discrete Time Linear Quadratic Differential Games
	40 Ganapathy Mani, Bharat Bhargava, Denis Ulybyshev, Jason Kobes and Puneet Goyal. Autonomous Aggregate Data Analytics in Untrusted Cloud
	31 Ryohei Yokobayashi and Takao Miura. Multidimensional Data Mining Based on Tensor Model

AIKE2018 WORKSHOPS

ABC 2018

Chair: Yunmook Nah, Dankook University, Korea

1. Jeong-Min Seo, Haanju Yoo, Kimin Yun, Hyunil Kim, Sang-Il Choi, *Behavior Recognition of a Person in a Daily Video Using Joint Position Information*
2. Sherzod Nurullayev and Sang-Woong Lee, *Parking occupancy detection for unseen dataset based on Dilated Convolutional Neural Networks*
3. Jaemin Jung, Myeonggeon Lee, ..., Seong-je Cho, Kyungwon Suh, *Android Malware Detection based on Useful API Calls and Machine Learning*
4. *Emmanuel Oyekanlu, Distributed Osmotic Computing Approach to Implementation of Explainable Predictive Deep Learning at Industrial IoT Network Edges with Real-Time Adaptive Wavelet Graphs*
5. Axel Gedeon Mengara, Jiwan Lee, Bonghee Hong, Woo Chan Kim(Agency for Defense Development), *Generation of data set for tactical moving objects*
6. Donghan Sun and Soochan Hwang, *DSSP: Stream Split Processing Model for High Correctness of Out-of-Order Data Processing*
7. SungHwan Jeon, Wonseok Choi, HaeJin Chung, Jin Taek Kim, Jonghoon Chun, and Yunmook Nah, *MapReduce Tuning to Improve Distributed Machine Learning Performance*
8. Woojin Jeon, Daeseon Choi and Ki-Woong Park, *Applying Time-Lapse Concepts into Storage System for Long-Term System Trace Analysis*
9. Chan Cheong, Kee-Won Kim and Seung-Hoon Kim, *Design of the Platform Architecture Providing Information for Barrier-free Tourism*
10. Ngoc-Quang Nguyen and Sang-Woong Lee, *Colorectal Segmentation Using Multiple Encoder-Decoder Network in Colonoscopy Images*

AI3D 2018

Chair: Salman Ahmed Shaikh, AIRC, AIST, Japan

1. Rocio Torres, Piero Fraternali, Federico Milani and Darian Frajberg, *A Deep Learning model for identifying mountain summits in Digital Elevation Model data*

2. Salman Shaikh, Akiyoshi Matono and Kyoung-Sook Kim, *Continuous Querying over Mobile Mapping Stream*

3. Jun Lee, Mikyoung Seo, Jinhwan Kim, Soyoung Hwang, Taehoon Kim and Kyoung-Sook Kim, *Management of Subdivided Dynamic Indoor Environments by Autonomous Scanning System*

4. Taehoon Kim, Jun Lee, Akiyoshi Matono and Kyoung-Sook Kim, *DOTLOOM: A Distributed Computing Platform for Three-dimensional Point Cloud Data*

5. Chen-Yu Hao, Mei-Hsin Chen, Tien-Yin Chou and Chia-Wei Lin, *The Research on the Framework Design for Integrating the Point Cloud Semantic Annotation Process and PointNET Classification Module*

IDP 2018

Chair: Kristen Yuanxi Li, Hong Kong Baptist University, Hong Kong

(1) Paolo Mengoni, Alfredo Milani, Yuanxi Li, Multi-term, *Semantic Context Elicitation from Collaborative Networks*

(2) Valentina Franzoni, Michele Lepri, Yuanxi Li, Alfredo Milani, *Efficient Graph-based Author Disambiguation by Topological Similarity in DBLP*

(3) Nikki L. J. Kuang, Clement H. C. Leung, Vienne W. K. Sung, *Stochastic Reinforcement Learning*

(4) James Jie Deng, Clement H. C. Leung, Yuanxi Li, *Emotion Recognition from Human Behaviors using Attention Model*

(5) Sheung Wai Chan, Valentina Franzoni, Paolo Mengoni, Alfredo Milani,
Yiu Ming Cheung, Context-based Image Semantic Similarity for Prosthetic Knowledge

(6) Haolan Zhang, Jiming Liu, Rao Kotagiri, Yun Xue*, A Hybrid Sampling Algorithm for Large Volume Data Processing*